

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

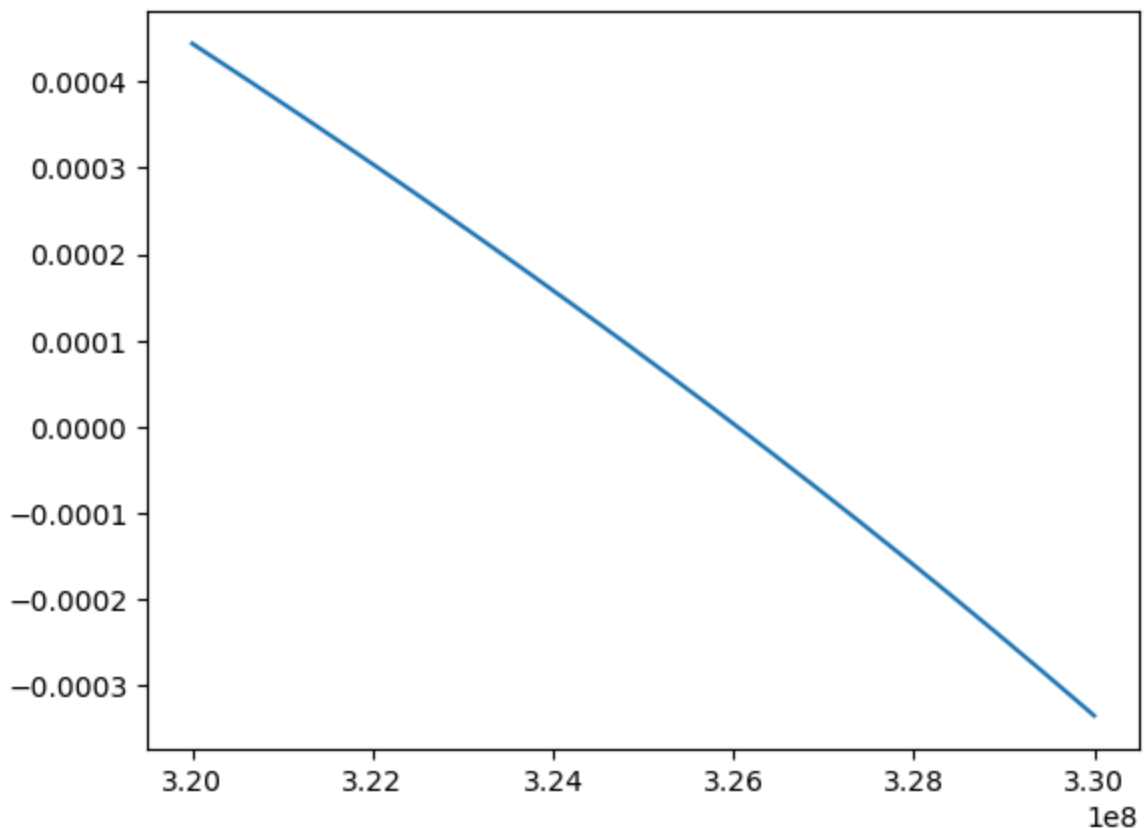
```
In [2]: G = 6.674*10**-11
M = 5.974*10**24
m = 7.348*10**22
R = 3.844*10**8
w = 2.662*10**-6
```

```
In [3]: def f(x):
return (G*M/x**2 - G*m/(R-x)**2 - w**2*x)
```

after a long time of zooming in i finally found it...

```
In [17]: fig, ax = plt.subplots()
x_ = np.linspace(3.2e8, 3.3e8, 1000)
ax.plot(x_, f(x_))
```

```
Out[17]: [<matplotlib.lines.Line2D at 0x77ff4390b620>]
```



```
In [4]: def secant(f, guess, delta=1, tolerance = 2**-32):
x0 = guess
x1 = x0 + delta
n = 0
while abs(f(x1))>tolerance:
x2 = x1 - (x1-x0)/(f(x1)-f(x0))*f(x1)
```

```
    x0 = x1
    x1 = x2
    n += 1
    return(x1, n)
```

In [5]: `secant(f, 1000000, 100)`

Out[5]: (326045071.6618108, 26)

In []: